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Comparison of Deep Learning and Gradient Boosting: ANN Versus XGBoost for Climate-Based Dengue Prediction in Bangladesh

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ABSTRACT

Background and Aims: Bangladesh is confronting an unprecedented threat from dengue, worsened by climate change. Understanding its trends and pinpointing climatic risk factors is vital for effective intervention and control. This study aimed to compare the artificial neural network (ANN) and eXtreme gradient boosting (XGBoost) models to identify the best model for predicting dengue incidence and determining key climatic risk factors in Bangladesh.

Methods: Monthly dengue data were obtained from the Directorate General of Health Services, while climate data were collected from the NASA website. Exploratory data analysis, as well as ANN and XGBoost models, were performed to analyze the data.

Results: From 2000 to 2023, Bangladesh reported 565,890 dengue cases, with 2023 having the highest outbreak at 79,598 cases. Precipitation, ranging from 0 mm to 1012.50 mm (mean = 181.58 mm), showed significant variability, with the highest recorded in August 2017. Relative humidity had a mean of 75.99%, and wind speed averaged 10.11 m/s. XGBoost outperformed other models, achieving the lowest root mean square error (918.83) and mean absolute error (479.44). The model identified precipitation as the key factor influencing dengue incidence, followed by relative humidity and wind speed.

Conclusion: This study's findings, which incorporate climate data and predictive modeling, offer valuable insights for preventing and controlling dengue epidemics and for early warning systems in Bangladesh. To mitigate the impact of future outbreaks, key factors such as precipitation, relative humidity, and wind speed should be closely monitored, alongside proactive measures and enhanced surveillance.

1 | Introduction

Dengue/Severe Dengue Fever (DF/SDF) is among the fastest-growing arboviral diseases in tropical regions like Bangladesh, and no widely recognized cure or vaccine is available. Over the past four decades, the rapid spread of the dengue virus (DENV) and its mosquito vectors, primarily *Aedes aegypti* and *Aedes albopictus*, has created a significant public health threat in tropical regions [1]. DF is considered a threat to 40% of the global population, with 390 million cases reported each year, and it is recognized as one of the top ten global health risks by the World Health Organization (WHO) [2]. As of April 30, 2024, the WHO has reported over 7.6 million dengue cases for the year, including 3.4 million confirmed cases, more than 16,000 severe cases, and over 3000 deaths [3].

The first recorded dengue epidemic in Bangladesh occurred in 1964, followed by a major outbreak in 2000, which saw 5551 reported cases and 93 fatalities [4, 5]. Since then, dengue has become endemic in Bangladesh. In 2018, the country reported

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over 10,000 instances of dengue for the first time. Remarkably, in 2019, Bangladesh experienced one of its largest dengue epidemics, with 101,354 cases and 164 dengue-related deaths [6]. In contrast to the severe outbreaks in previous years, Bangladesh reported only 1405 dengue cases and three confirmed dengue-related deaths in 2020. In 2023, dengue seems set to surpass all previous records. The latest statistics show a rising trend in reported dengue cases, with a total of 316,160 cases, including 1705 fatalities [7]. These figures demonstrate how terrible the dengue issue has become in Bangladesh.

Bangladesh faces a significant risk of contracting dengue due to its tropical monsoon climate, which creates conditions that strongly favor dengue transmission [8]. Specifically, the monsoon season (usually May to September) brings high temperatures [9], high humidity, and heavy rainfall to the nation [10, 11]. These weather patterns cause urban areas to become waterlogged and accumulate standing water, making them ideal breeding grounds for Aedes mosquitoes. Additionally, the economic burden of dengue is also considerable. A prior study conducted in Bangladesh revealed that the average cost to society for a dengue episode was $ 479.02, with the sum ranging between $ 341.67 for patients treated at public hospitals and $ 567.12 for those treated in private hospitals [12]. The households' out-of-pocket expenses accounted for a larger share of the total costs of illness (66%), with the cost burden being significantly higher for the poorest households compared to the wealthiest. These economic challenges underscore the significant financial burden that dengue places on individuals and healthcare systems, highlighting the necessity for effective predictive models to direct timely interventions and lessen the disease's impact.

With the increasing prevalence of dengue in Bangladesh, timely treatment relies on accurate predictive models. According to the World Health Organization (WHO), monitoring, early warning systems, and climate-based risk assessments are essential components of dengue management programs aimed at preventing outbreaks [13]. This study also advances Sustainable Development Goal (SDG) 3, which seeks to enhance global health outcomes by strengthening health systems and preventing communicable diseases [14]. Therefore, a comprehensive investigation using long-term data is essential to understand dengue outbreak trends and identify the climatic factors that significantly impact its spread in Bangladesh. Several earlier studies have utilized methods like time series analysis [1], generalized linear model [15], negative binomial regression [16], Poisson regression [8, 17], and machine learning models [18, 19] to investigate the association between dengue incidence and climatic factors. While these studies provided valuable insights, they were limited in managing complex, nonlinear, and high-dimensional data. However, no comparative study using long-term data has yet been conducted to determine which model best predicts dengue outbreaks and identifies the most influential climatic factors driving the disease's incidence in Bangladesh. A comparative analysis is crucial to determine the most reliable model for predicting dengue outbreaks and accurately assessing the key climatic drivers. Recent advancements in dengue modeling in Bangladesh including Stochastic Bayesian Downscaling (SBD) [20] and SIS vector–host model [21] also highlight the significance of integrating complex data to improve forecasting and control strategies. In contrast, the artificial neural network (ANN) is a deep learning model excellent for handling nonlinear data and has outperformed in many sectors, such as COVID-19 [22], malaria [23], influenza [24], tuberculosis [25], Diabetes [26] and so on. On the other hand, eXtreme gradient boosting (XGBoost) is a tree-based ensemble machine learning model known for handling nonlinear data and providing greater prediction accuracy, having outperformed in various fields [9, 27–29]. Therefore, this study compares deep learning with machine learning to identify the best model for predicting dengue incidence in Bangladesh. The findings will help inform public health decision-making and enhance early warning systems for dengue outbreaks.

2 | Methods

2.1 | Data Source

This study compiled a data set of monthly dengue cases along with seven climatic factors—maximum, minimum, and mean temperature, relative humidity, surface pressure, maximum wind speed, and precipitation. Dengue cases data were obtained from the Directorate General of Health Services (DGHS) from January 2000 to December 2023 [30], while climate data were sourced from the NASA Langley Research Center (LaRC) website from January 2000 to December 2022 [31]. Since the climatic data were collected between 2000 and 2022, their dimensions do not align with the dengue data which could cause issues with modeling. To circumvent this problem, dengue cases data for 2023 were removed and the remaining dengue data were matched with the climatic data. More details of the data are presented in Table 1.

2.2 | Statistical Analysis

The statistical analysis began with a descriptive examination of the data. To ensure that the climatic variables were on similar scales, which is essential for model performance, preprocessing techniques such as scaling were applied. Specifically, the climatic variables were adjusted to a [0, 1] range using the

TABLE 1 | Description of all input features.

Code	Description
y	Monthly dengue cases
x1	MERRA-2 temperature at 2 meters (C)
x2	MERRA-2 temperature at 2 meters minimum (C)
x3	MERRA-2 temperature at 2 meters maximum (C)
x4	MERRA-2 relative humidity at 2 meters (%)
x5	MERRA-2 precipitation corrected (sum)
x6	MERRA-2 surface pressure (kPa)
x7	MERRA-2 wind speed at 50 meters maximum (m/s)

Abbreviation: MERRA-2, modern-era retrospective analysis for research and applications, version 2.

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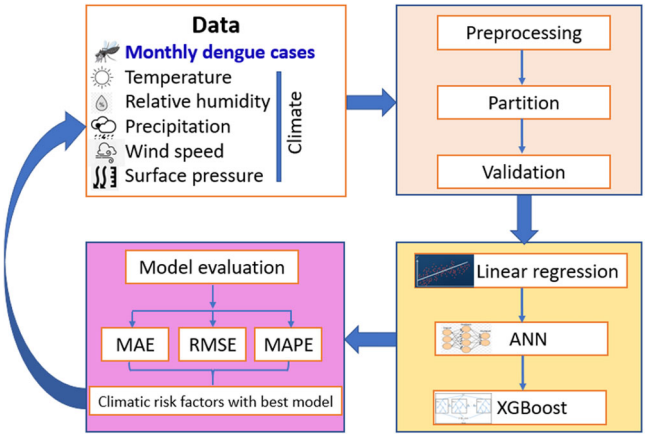


FIGURE 1 | Summary of the proposed study design, highlighting its key components.

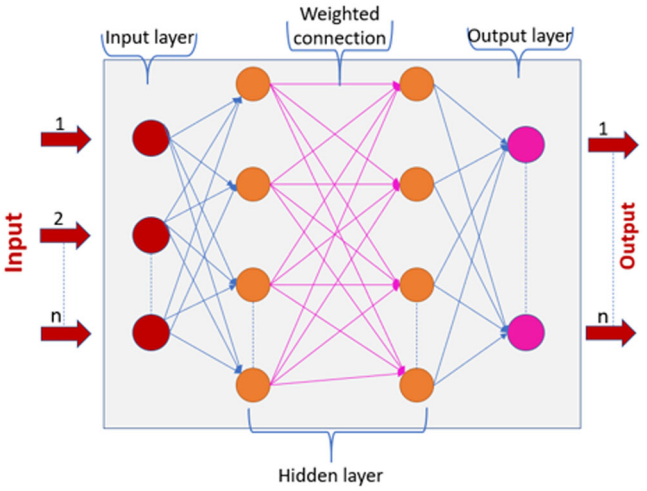


FIGURE 2 | The overall structure of an artificial neural network (ANN) configuration.

min-max scaling method. This scaling prevents variables with larger ranges from disproportionately affecting the model, ensuring that each variable contributes equally to the analysis. Before splitting the data, multicollinearity was assessed using the Variance Inflation Factor (VIF) [32]. The data set was then divided into training (80%) and test (20%) sets. The training set was used to develop the models (Figure 1). To ensure model robustness, 10-fold cross-validation was performed, and multiple hyperparameters were tuned. All analyses were performed in RStudio (version 4.4.0) [33] on a 64-bit Windows 11 operating system with an AMD Ryzen 3 processor and 4 GB of RAM. Descriptive statistics were computed using the "dplyr," "car," "lubridate," and "tidyr" packages, while the model building was carried out using "caret," "MLmetrics," and "dplyr." Data visualization was performed using the "ggplot2" and "ggcorrplot" packages.

2.3 | Artificial Neural Network

ANNs are computer models that are modeled after the composition and operations of the human brain [34]. They are made up of hierarchically processed artificial neurons arranged in linked layers. It comprises three layers including the input layer, the hidden or calculation layer, and the output layer (Figure 2) [35, 36]. In an ANN, neurons are connected by weights that facilitate the transfer of information between nodes during the computation process [37]. These weights are adjusted during training, allowing the computation process to guide previous nodes toward a common next node, ensuring the continuation of processing [38]. Each layer contains neurons and performs distinct functions. The input layer converts environmental data, images, or signals into normalized components suitable for mathematical computations in the output layer. The computation takes place in the hidden layer(s), which contain the largest number of neurons and perform calculations through effective neuronal communication. Finally, the neurons in the output layer present the computed results to the user [39–41].

In this study, an ANN was applied to predict the incidence of dengue based on climatic predictors. The network was trained using the training data set, where the input layer received the standardized climatic data as input features.

2.4 | eXtreme Gradient Boosting

The XGBoost, a type of boosting algorithm that combines multiple learning models to enhance predictive performance beyond any individual model, has demonstrated excellence in various fields [42]. It was first introduced in 2011 by Chen Tianqi and Carlos Gestrin, and later, several researchers contributed to its refinement and enhancement in subsequent studies [9]. It consists of two components: the decision tree algorithm and the gradient boosting algorithm. Gradient boosting is a powerful method for building predictive models and serves as a key approach to boosting. The core principle of boosting involves sequentially creating weak learners and progressively combining them to enhance overall performance [43]. Because of its effectiveness, scalability, and excellent prediction accuracy, XGBoost has become one of the most widely used machine learning algorithms. The approach uses regularization techniques to prevent overfitting and ensure strong generalization, intending to optimize both model accuracy and computational efficiency [44]. The objective function of the model is

$$Obj^{(t)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) + constant. \quad (1)$$

In this model, y_i represents the observed dengue case counts, while $\hat{y}_i^{(t-1)}$ represents the predicted value from the previous iteration. The input vector x_i consists of climate variables used as predictors, n denotes the number of observations, f_t represents the function trained by the algorithm. The regularization term $\Omega(f_t)$ helps prevent overfitting. The loss function l quantifies the deviation between the actual label and the estimate from the previous stage, based on the output of the new tree [45]. The XGBoost model is hyperparameter sensitive, and tuning parameters can significantly improve its performance [44]. In this study, several hyperparameters like nround (number of boosting rounds), max_depth (maximum depth of trees to control overfitting), eta (learning rate), gamma (minimum loss reduction for a split), colsample_bytree (fraction of features used per tree), min_child_weight (minimum instance weight per leaf), and subsample (fraction of samples

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used per tree) were tuned to optimize the XGBoost model's performance and enhance accuracy and generalization.

2.5 | Model Validation and Evaluation

Using the training data, the models were constructed, and their performance was improved through cross-validation and hyperparameter tuning. Specifically, this study applied 10-fold cross-validation, which divides the data into 10 groups and repeatedly trains the model on nine of them while testing on the remaining one [44]. This approach helps reduce both overfitting and underfitting.

The actual accuracy of a model can be determined by comparing its predicted values with the observed data. To assess its performance, a range of evaluation metrics can be applied to measure accuracy effectively [46]. This study evaluates the predictive performance of the model using three widely used forecasting metrics: mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE). MAE represents the arithmetic mean of the absolute differences between predicted and actual values, providing the average absolute forecasting error [45]. RMSE, a commonly used metric, measures the differences between predicted and observed values by taking the square root of the average squared error. MAPE expresses accuracy as a percentage by calculating the cumulative absolute percentage error for each time frame, obtained by dividing the absolute difference between actual and predicted values by the actual values. This metric represents the mean error in percentage terms. The mathematical expressions for these metrics are presented below:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | \hat{y}_i - y_i |. \quad (2)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (\hat{y}_i - y_i)^2}. \quad (3)$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n} \left| \frac{\hat{y}_i - y_i}{y_i} \right| \times 100\%. \quad (4)$$

In this context, n represents the total number of observations, $\hat{y}_i$ denotes the predicted value, y_i signifies the actual value, and $\hat{y}_i - y_i$ indicates the residual [27].

3 | Results

Dengue cases in Bangladesh varied significantly between 2000 and 2023, with a total of 565,890 confirmed cases. The highest number of cases occurred in 2023, with a mean of 26,764.92 (SD = 31,932.96) and a maximum of 79,598 cases in a single month (Table 2), indicating a severe outbreak (Supporting Information: Figure S1). In contrast, 2014 recorded the lowest peak, with a mean of 31.25 (SD = 33.27) and a maximum of just 82 cases. From 2000 to 2016, dengue cases remained relatively low, with several years having months with zero reported cases. However, since 2017, there has been a sharp increase in cases, with major outbreaks in 2019 (Table 2).

TABLE 2 | Descriptive statistics of dengue incidences in Bangladesh from 2000 to 2023.

Year	Min	Max	Median	Mean ± SD
2000	0	2290	93	462.58 ± 684.63
2001	0	655	88	202.50 ± 255.76
2002	0	3281	5	519.33 ± 1004.11
2003	0	372	1.50	40.50 ± 108.19
2004	0	1261	97	327.83 ± 460.57
2005	0	337	2.50	87.33 ± 132.56
2006	0	972	46	183.33 ± 314.48
2007	0	179	0	38.83 ± 66.92
2008	0	475	0	96.08 ± 161.20
2009	0	188	0	39.50 ± 71.66
2010	0	183	0	34.08 ± 66.38
2011	0	800	0	113.25 ± 233.49
2012	0	262	8	55.92 ± 81.71
2013	0	495	51	145.75 ± 177.14
2014	0	82	13	31.25 ± 33.27
2015	0	965	51.50	263.50 ± 375.09
2016	3	1544	199.50	505.00 ± 588.63
2017	36	512	200.50	230.75 ± 165.27
2018	7	3087	294	845.67 ± 1064.10
2019	17	52636	1565.50	8443.67 ± 15226.60
2020	10	546	46	99.42 ± 148.17
2021	3	7841	739.50	2369.08 ± 3054.99
2022	20	21932	1154	5257.42 ± 8157.92
2023	111	79598	7622	26764.92 ± 31932.96

Abbreviations: Max, maximum; Min, minimum; SD, standard deviation.

Figure 3 illustrates the monthly distribution of dengue cases in Bangladesh from 2000 to 2023 where a noticeable increase in cases starting around June, with August having the highest number, followed by September and October. This pattern suggests that dengue cases tend to rise toward the end of the monsoon season.

From 2000 to 2022, the mean temperature ranged from 15.55°C to 31.90°C, with an average of 25.52°C (Figure 4). The highest mean temperature (31.90°C) was recorded in April 2014, while the lowest (15.55°C) occurred in January 2018 (Supporting Information: Table S1). Monthly precipitation varied significantly, ranging from 0 to 1012.50 mm, with a mean of 181.58 mm. Precipitation began rising in March, peaking in July and gradually decreased from September (Figure 4). The highest recorded precipitation was 1012.50 mm in August 2017 (Supporting Information: Table S1). Other variables, such as average relative humidity (75.99%) and wind speed (10.11 m/s), also showed considerable variability (Supporting Information: Table S1).

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Before modeling the data, a Pearson bivariate correlation analysis was conducted to explore the relationship between climatic factors and dengue cases. The results indicated a positive association between dengue cases, relative humidity, and precipitation (Figure 5). However, the correlation matrix suggested potential multicollinearity, particularly among mean, minimum, and maximum temperatures. To address this, a VIF test was performed, a widely used method for detecting multicollinearity. A VIF value greater than 10 indicates severe multicollinearity [47], and the test confirmed that mean, minimum, and maximum temperatures had excessively high VIF values (Supporting Information: Table S2).

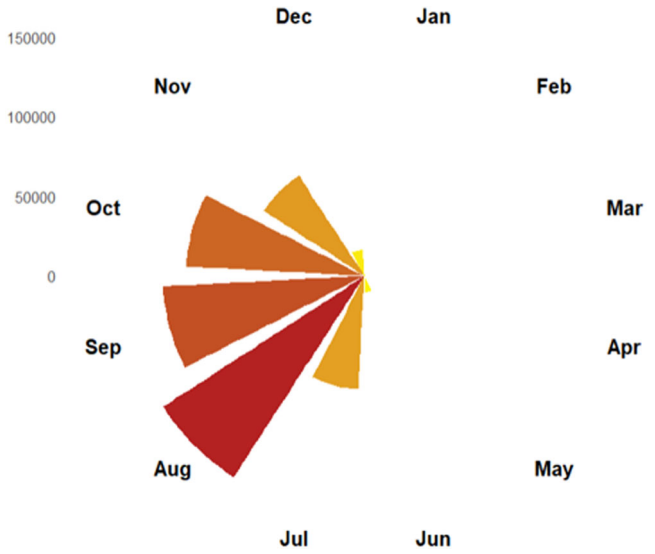


FIGURE 3 | Monthly distribution of dengue cases in Bangladesh from 2000 to 2023.

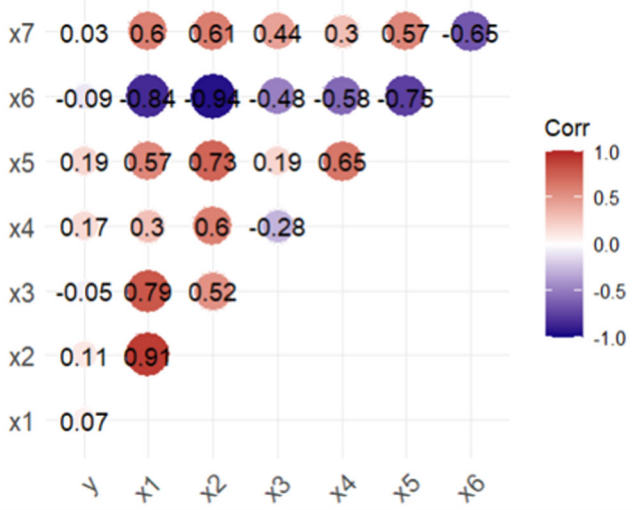


FIGURE 5 | Pearson bivariate correlation between dengue cases and climatic factors.

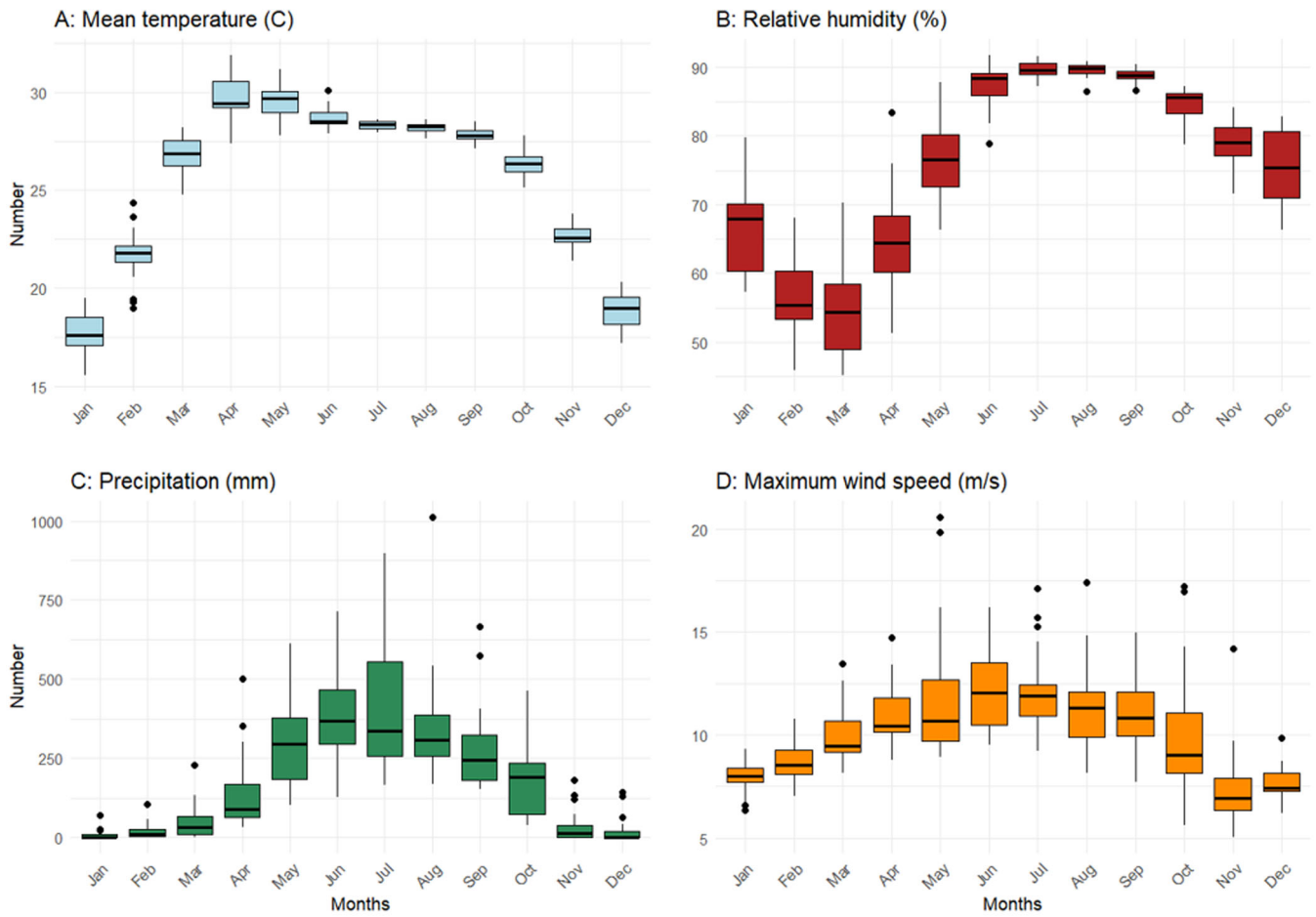


FIGURE 4 | Monthly distribution of climatic factors in Bangladesh from 2000 to 2022.

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To address multicollinearity, regularization techniques such as Ridge or Lasso regression [48] and dimensionality reduction methods such as principal component analysis (PCA) [49] act as alternatives to variable exclusion. While regularization techniques impose penalties on the coefficients of correlated variables to shrink them toward zero, PCA can transform correlated variables into a set of uncorrelated principal components. However, for the following reasons, variable exclusion was favored over these options in this study: the correlation analysis indicated that the temperature variables were insignificant, and their inclusion would likely result in overfitting or problems with interpretability. Furthermore, excluding these temperature variables contributed to maintaining model robustness without unnecessary complexity. The final models included the remaining four climatic factors as covariates, which were more directly related to dengue incidence and less susceptible to multicollinearity.

This study trained the models using the training data set, with linear regression as the baseline to explore linear relationships. The ANN model was optimized through a preprocessing technique, including scaling and normalization. Key hyperparameters decay and size = 5, were tuned to enhance performance. The decay parameter regularizes the model, preventing overfitting, while size = 5 optimizes the number of neurons in the hidden layer, balancing complexity and prediction accuracy. These adjustments were crucial for improving the model's performance. On the other hand, the XGBoost model was developed by tuning various hyperparameters and using cross-validation to improve performance. The max_depth was set to 3 to prevent overfitting while maintaining model interpretability, as deeper trees can lead to excessive variance. Other hyperparameters, including nround = 100, eta, gamma, colsample_bytree, min_child_weight, and subsample, were tuned using cross-validation to optimize accuracy and ensure robust predictions (Supporting Information: Table S3 and Figure S2). After training all the models, predictions were made on the test set, and their performance was evaluated.

In terms of prediction accuracy, XGBoost outperformed the other two models, achieving the lowest RMSE (918.83) and MAE (479.44) on the test data set. While ANN had the highest test error (RMSE = 2229.66), indicating lower generalization, Linear Regression (LR) showed moderate performance (RMSE = 1444.42). Despite its strong performance on the training data, the ANN's higher test errors suggest potential overfitting. Overall, XGBoost emerged as the most accurate model for forecasting dengue cases in this study (Table 3).

TABLE 3 | Evaluation metrics of the fitted models.

Models	Data set	RMSE	MAE	MAPE
LR	Training	4314.93	1546.48	1.18
	Test	1444.42	1083.67	1.02
ANN	Training	4005.70	1441.92	1.54
	Test	2229.66	1260.98	1.92
XGBoost	Training	3261.38	945.13	1.55
	Test	**918.83**	**479.44**	2.25

Abbreviations: ANN, artificial neural network; LR, linear regression; MAE, mean absolute error; MAPE, mean absolute percentage error; RMSE, root mean square error; XGBoost, eXtreme gradient boosting.

The XGBoost model, which outperformed the other two models, identified precipitation as the most significant predictor of dengue cases, followed by relative humidity and maximum wind speed. This ranking suggests that while humidity and wind speed had some influence, precipitation had the strongest impact (Figure 6).

4 | Discussion

Dengue, a potentially deadly mosquito-borne disease, has experienced a substantial rise in Bangladesh, a country located in South Asia, in recent years. This study identified the highest peaks of dengue in 2019 and 2023, although 2023 was excluded from the analysis due to the alignment of the timeframe with climatic factors. Notably, 2019 saw the largest epidemic before the unprecedented outbreak in 2023. Monthly dengue distribution showed a sharp rise from June, peaking in August, followed by September and October, indicating an increase toward the end of the monsoon season. Climatic variability, including fluctuations in humidity, precipitation, and temperature, plays a crucial role in the DENV transmission cycle [50]. The main carriers of dengue transmission in Bangladesh are Aedes mosquitoes, particularly *Aedes aegypti* and *Aedes albopictus* [51]. The ecology of these insects is significantly influenced by human and environmental factors. Urbanization, rapid population growth, and inadequate drainage systems have created ideal breeding conditions for these mosquitoes in urban areas. Many urban and peri-urban locations have stagnant water in open containers, poor water storage practices, and poorly maintained water infrastructure, all of which provide ample breeding grounds. Increased rainfall during the monsoon season [52] floods these containers and saturated areas, worsening these breeding opportunities. Thus, this study aimed to compare the performance of an ANN and XGBoost to identify the best model for predicting dengue incidence and determining the most influential climatic factors in Bangladesh.

ANN, a deep learning model, effectively handles both linear and nonlinear relationships, producing highly accurate outputs [53].

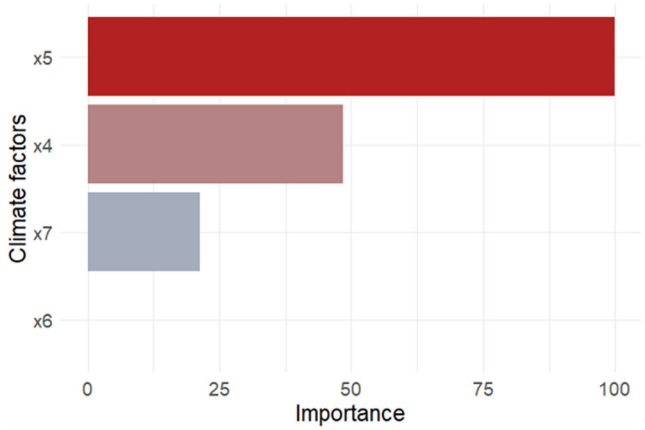


FIGURE 6 | The key climatic factors identified by the XGBoost model that influence dengue incidence in Bangladesh, x4: relative humidity; x5: precipitation; x7: maximum wind speed.

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However, a drawback of ANN models is that they are often referred to as "black boxes" due to the intricate relationships between layers and neurons, making it challenging to understand how the model arrived at a particular conclusion [22]. In contrast, XGBoost, a decision tree-based ensemble model, offers greater interpretability through feature selection, transparency, and the ability to capture complex nonlinear patterns [54, 55]. Before modeling the data, this study performed Pearson bivariate correlation to examine the relationship between dengue cases and climate factors. The correlation analysis revealed a positive relationship between precipitation, relative humidity, and dengue cases. Temperature was initially considered as a predictor but was excluded from the analysis due to multicollinearity with other climate variables, particularly relative humidity, and precipitation, which could have distorted the model's performance. The LR, ANN, and XGBoost models were then fitted with climate factors as predictors and compared in terms of performance, with linear regression serving as the baseline model. Evaluation using RMSE, MAE, and MAPE confirmed that XGBoost outperformed the other models.

Using the best-performing XGBoost model, the study identified the most influential climatic factors. Monthly precipitation emerged as the strongest predictor of dengue incidence, aligning with previous studies [8, 56]. Heavy rainfall likely contributes to dengue outbreaks by creating stagnant water pools, ideal breeding sites for Aedes mosquitoes, the primary dengue vectors [57]. However, a prior study claimed that precipitation was negatively associated with dengue incidences [8]. By identifying precipitation as the strongest predictor, this model enables health authorities to prioritize larvicidal treatments in regions with rainfall above 180 mm/month. Relative humidity was also found to significantly impact dengue incidence, supporting findings from earlier studies [58, 59]. Studies also revealed that humidity $\geq 56\%$ was strongly associated with dengue incidence [60]. This may be because high humidity enhances *Aedes aegypti* survival and reproduction, thereby increasing transmission risk. Additionally, maximum wind speed at 50 meters was found to influence dengue incidence, which aligns with previous research [61]. Wind conditions can affect the movement and dispersal of Aedes mosquitoes, impacting transmission patterns.

Recent research in low- and middle-income countries shows that machine learning can enhance dengue monitoring and public health decision-making [62]. For instance, an ensemble machine learning system accurately predicted dengue incidence rates across extensive administrative regions in Brazil and departments in Peru by incorporating earth observation data, climate models, and socioeconomic characteristics [63]. This approach produced strong results in forecasting dengue outbreaks with minimal uncertainty and demonstrated its applicability across various regions. Similarly, other research has emphasized the importance of integrating machine learning into dengue prediction models to enable prompt public health interventions, optimize resource allocation, and impact policy decisions [64, 65].

The results of this study have direct implications for public health. By incorporating XGBoost-based predictions into early warning systems, health officials can anticipate dengue outbreaks and implement vector control measures promptly. For example, suppose the model predicts high-risk months based on climatic trends. In that case, local governments can intensify mosquito control efforts and public awareness campaigns in advance, while healthcare facilities can allocate resources strategically to prepare for peak dengue seasons. Public health initiatives should be enhanced before high-risk seasons to raise community awareness and promote preventive measures such as mosquito control and personal protection. Improving model applicability through collaboration with health authorities like DGHS can ensure alignment with dengue monitoring and control efforts. Considering the increasing impact of climatic variability on dengue transmission, data-driven models like XGBoost can enhance disease monitoring and response strategies in Bangladesh, reducing dengue-related morbidity mortality.

5 | Limitation

A limitation of this study is the omission of the temperature variable due to multicollinearity with other environmental parameters, which may have reduced the model's ability to fully capture its effect on dengue incidence. This omission might induce biases because temperature has a well-established impact on mosquito ecology. Although this choice was necessary to preserve model stability, future studies could investigate interaction effects or delayed temperature variables to better account for its impact while reducing collinearity issues. Furthermore, socioeconomic factors, such as population density and access to sanitation, were not included in this study, which may have influenced the findings. Given that these characteristics are known to affect dengue transmission and the effectiveness of public health interventions, future models should consider them. Incorporating these variables would result in a more comprehensive understanding of the complex factors influencing dengue incidence. Additionally, potential data collection biases, such as underreporting of dengue cases, and the limitations of using artificial neural networks for time-series forecasting should be addressed in future studies by evaluating alternative models and strategies to mitigate these issues biases.

6 | Recommendation

To offer a more thorough understanding of dengue transmission, future research should consider including other factors such as temperature, human behavior, and vector control measures. Furthermore, incorporating socioeconomic—such as income levels, access to healthcare, and population density—and real-time monitoring could enhance early warning systems and public health, aligning with the need for enhanced surveillance. Combining epidemiological, environmental, and socioeconomic data to strengthen monitoring may result in more focused control efforts and more successful preventative tactics.

7 | Conclusion

This study emphasizes the critical role of climatic variables, particularly precipitation, in forecasting dengue incidence in

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Bangladesh. A comparison of the XGBoost and ANN models revealed that XGBoost outperforms ANN in terms of prediction accuracy. The findings underscore the importance of incorporating climate factors into dengue forecasting models. Climate-driven projections, such as precipitation thresholds, could help health authorities launch public awareness campaigns and mosquito control measures on time, especially during high-risk months. Persistent precipitation monitoring might act as an early warning system, enabling the timely adoption of vector control techniques, resource allocation, and healthcare readiness before dengue cases spike. By taking such preventive steps, public health responses to dengue epidemics could be improved, and early warning systems might become more effective.

Author Contributions

Arman Hossain Chowdhury: conceptualization, investigation, writing – original draft, methodology, validation, visualization, writing – review and editing, software, formal analysis, project administration, data curation, supervision, resources.

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

All data and source codes are available at https://github.com/arman2018/dengue-machine-deep-learning-prediction-bd.

Transparency Statement

The lead author Arman Hossain Chowdhury affirms that this article is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

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